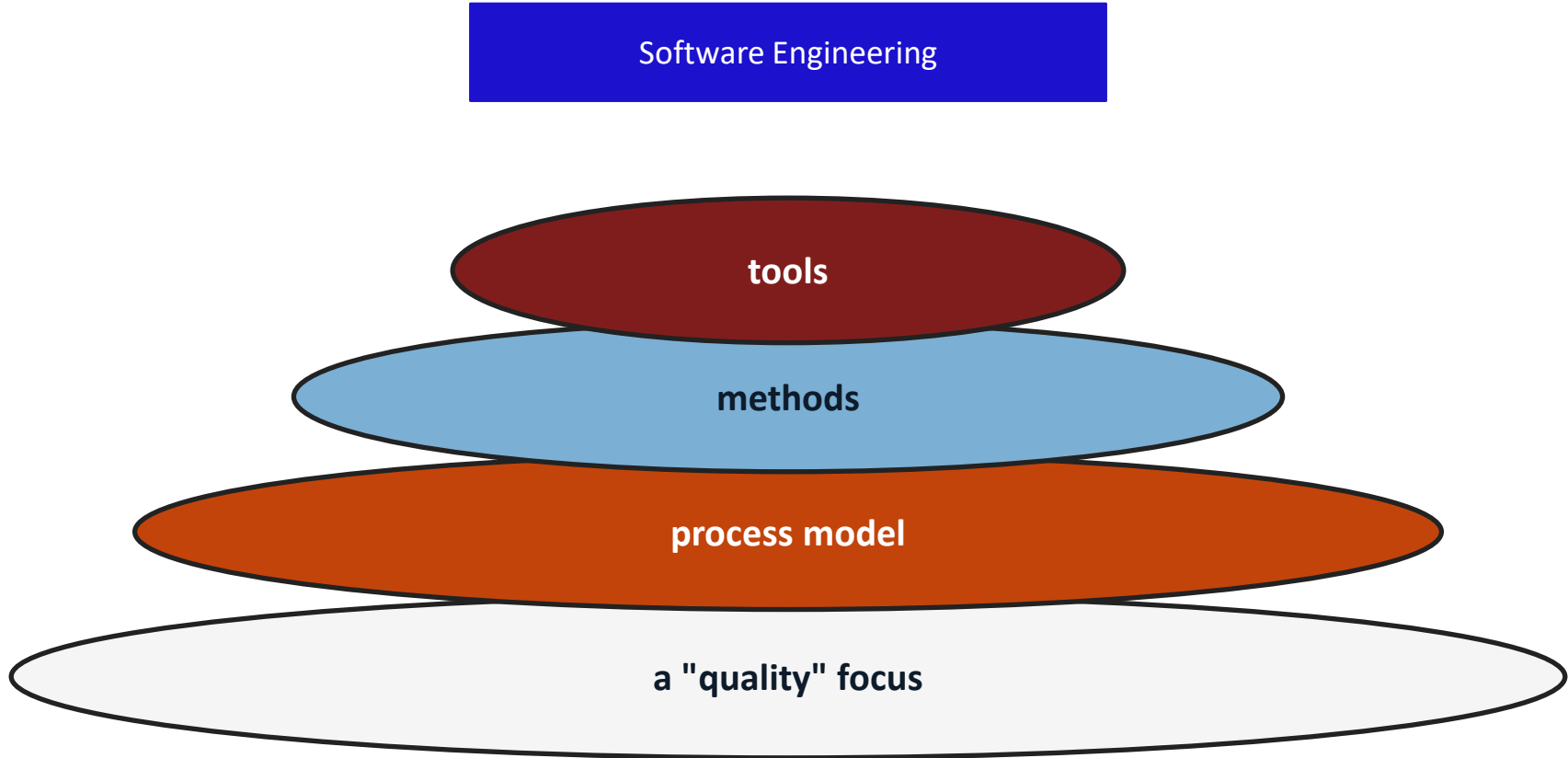


SOFTWARE QUALITY ASSURANCE

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SOFTWARE ENGINEERING: A Quality Focus



Quality is the foundation upon which all software engineering rests.

What is Quality?

"Quality is a characteristic or attribute of something. Attributes are measurable characteristics — things we can compare to known standards."

01 Quality of Design

Characteristics that designers specify for an item

Grade of materials, tolerances and performance specifications

Defines the intended quality level of a product

02 Quality of Conformance

Degree to which design specifications are followed during manufacturing

Greater conformance = higher quality level

Focused primarily on the implementation phase

SOFTWARE QUALITY

Software quality is defined as conformance to explicitly stated functional and performance requirements, explicitly documented development standards, and implicit characteristics expected of all professionally developed software.



D

Quality of Design in Software

Encompasses requirements, specifications, and the design of the system. Defines what the software should do and how well.



C

Quality of Conformance in Software

An issue focused primarily on implementation. If implementation follows design and system meets goals, conformance quality is high.



I

Implicit Characteristics

Expected traits of all professional software — maintainability, reliability, usability, and security — even if not explicitly stated.

QUALITY CONTROL

Quality control involves a series of inspections, reviews, and tests used throughout the software process to ensure each work product meets the requirements placed upon it.

Feedback Loop

Includes a feedback loop to the process that created the work product. Measurement and feedback allow tuning the process when products fail specifications.

Part of Manufacturing

Quality control is viewed as part of the manufacturing process — embedded into production, not separate from it.

Automated or Manual

QC activities may be fully automated, entirely manual, or a combination of automated tools and human interaction.

Scope

Applies to every work product at every phase: requirements docs, design specs, code, test plans, test results, and deliverables.

QUALITY ASSURANCE

Quality assurance consists of the auditing and reporting functions of management. Its goal is to provide management the data necessary to be informed about product quality, gaining insight and confidence that quality goals are being met.

How QA Differs from QC

Aspect	Quality Control (QC)	Quality Assurance (QA)
Nature	Operational	Managerial / Oversight
Focus	Product — finding defects	Process — preventing defects
Who does it	Testing/engineering team	SQA group + management
Output	Bug reports, test results	Audits, reports, data for management
Action on failure	Fix the product	Fix the process

COST OF QUALITY

Cost of quality includes all costs incurred in pursuit of quality or performing quality-related activities. It provides a baseline, identifies reduction opportunities, and enables comparison.

P

Prevention Costs

- Quality planning
- Formal technical reviews
- Test equipment
- Training

A

Appraisal Costs

- In-process & inter-process inspection
- Equipment calibration & maintenance
- Testing activities

F

Failure Costs

- Internal: rework, repair, failure mode analysis
- External: complaint resolution, product return
- Help line support, warranty work

Goal: Shift investment from failure costs toward prevention costs to improve overall quality and reduce total expenditure.

FAILURE COSTS: Internal vs External

Internal Failure Costs

Defects found BEFORE shipment

- Rework: correcting defective work
- Repair: fixing components that fail
- Failure mode analysis: diagnosing root cause

These are caught before the customer sees them.

External Failure Costs

Defects found AFTER shipment

- Complaint resolution
- Product return and replacement
- Help line support
- Warranty work

These are the most expensive — the customer is already affected.

Principle: Investing in prevention and appraisal dramatically reduces failure costs.

SQA ACTIVITIES

SQA is composed of tasks associated with two constituencies: **software engineers** who do technical work and an **SQA group** responsible for quality assurance planning, oversight, record keeping, analysis, and reporting.

Software Engineering Team

- Performs technical work according to defined processes
- Applies standards during design and coding
- Participates in formal technical reviews
- Implements corrections identified by SQA

SQA Group

- Assists software team in achieving high-quality products
- Plans and oversees quality assurance activities
- Audits work products for compliance
- Records noncompliances and reports to senior management

SQA ACTIVITIES — Plan, Process & Reviews

01

Prepare an SQA Plan

Developed during project planning and reviewed by all stakeholders. Governs all SQA activities for the project.

- Evaluations to be performed
- Audits and reviews to be performed
- Applicable standards
- Error reporting and tracking procedures
- Documents produced by SQA team
- Feedback to software project team

02

Participate in Process Description

The SQA group reviews the selected process for compliance with organizational policy, internal/external standards, and the project plan.

- Reviews chosen software process
- Checks internal standards compliance
- Checks externally imposed standards

03

Review Software Engineering Activities

The SQA group identifies, documents and tracks deviations from the defined process, and verifies that corrections are made.

- Verifies compliance with defined process
- Documents deviations
- Tracks corrections

SQA ACTIVITIES — Audits, Records & Reporting

04

Audit Software Work Products

The SQA group reviews selected work products, identifies and documents deviations, verifies corrections are made, and periodically reports results to the project manager.

05

Ensure Documented Procedures for Deviations

Ensures that deviations in software work and work products are documented and handled according to documented procedures.

06

Record & Report Noncompliance

Records any noncompliance found during audits and reviews, and escalates reports to senior management when issues are unresolved.

Escalation Path: SQA Group → Project Manager → Senior Management. Non-compliance must be formally tracked until resolved.

QUALITY CONCEPTS : Three Levels of Concern

Quality management operates at three distinct levels, each with a different scope and responsibility.

Organizational Level

Establish a framework of organizational processes and standards that lead to high-quality software across all projects and teams.

Process Level

Apply specific quality processes and check that planned processes have been followed. Involves defined workflows, reviews, and audits.

Project Level

Establish a quality plan for the project. Define quality goals, processes to be used, and standards applicable to the specific project.

Broader scope at top → narrower, project-specific at base. Each level feeds quality into the level below.

STATISTICAL SOFTWARE QUALITY ASSURANCE

Statistical SQA involves collecting, categorizing, and analyzing defect data to drive systematic quality improvements using statistical principles.

1

Collect & Categorize

Gather information about software defects. Categorize by type, phase of detection, severity, and cause.

2

Trace to Root Cause

Attempt to trace each defect to its underlying cause: design error, nonconformance to spec, standards violation, poor communication.

3

Apply Pareto Principle

80% of defects can be traced to 20% of all possible causes. Isolate the "vital few" — the 20% that cause most problems.

4

Correct the Vital Few

Once the vital few causes are identified, apply resources to correct the problems that caused the majority of defects.

Pareto Principle: 80% of defects come from just 20% of causes — fix those causes first.

Thank You

